

WestQuant Hamiltonian Representation Explorer (Project B)

Slide 1 — Project Overview

Problem: A mathematical optimization problem does not determine a unique useful Hamiltonian representation. Penalty strength, variable encoding, and landscape all matter.

Approach: Generate multiple valid representations (scaling, permutation, bit-complement, MWIS penalty family) and verify equivalence by exhaustive state-by-state comparison. Each is classified: EXACT_EQUIVALENT, GROUND_STATE_EQUIVALENT, SAME_PROBLEM_DIFFERENT_DYNAMICS, APPROXIMATE, or INVALID.

Results: For MWIS, penalty strengths U in $[5.5, 8.9]$ all yield GROUND_STATE_EQUIVALENT Hamiltonians with the same optimum but different gaps (1.0 to 6.9). Bit-complement flips interaction signs, making a Hamiltonian non-native-Rydberg.

Importance: The Hamiltonian representation choice drives 62% of solution-probability variance in our factorial experiment, vs 10% from embedding. Representation itself is an optimization variable.

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Slide 2 — QoolQit Experience

Positive: QoolQit's `DataGraph.from_matrix` and `interaction_matrix()` made it easy to connect logical Hamiltonians to physical interactions. The $1/r^6$ Rydberg convention is natural for QUBO/MWIS. `Register.from_coordinates` enables rapid coordinate experimentation.

Challenges: Native Rydberg interactions are repulsive ($J > 0$), so Hamiltonians with negative pair interactions (e.g., after bit-complement) are not directly native-realizable. The realizability classifier must flag this before expensive embedding. DMM is needed for MWIS node weights.

Workflow: The five-class equivalence taxonomy forced mathematical rigor. Exhaustive verification for $n \leq 16$ ensured no false equivalence claims. The realizability classifier prevents wasted computation on non-native Hamiltonians.